

AI

## AI with Rav

Learn AI the way you actually think.



# MACHINE LEARNING

## in 30 Days

DAY 1 · VIDEO 1

What is Machine Learning, really?

# What is Machine Learning, really?

**In One Line:** Machine Learning is teaching a computer to learn patterns from examples — instead of you writing the rules by hand.

## Start here: the chai stall

Picture **Ramesh**, who runs a chai stall outside a Mumbai station.

Nobody gave Ramesh a rulebook. But after 3 years, he *just knows*:

- Rainy evening + train delayed → **sell more chai, keep extra ginger**
- Hot afternoon → **nimbu paani sells, chai doesn't**
- Salary day (1st) → **everyone buys, stock up**

Ramesh was never *programmed*. He **learned from thousands of days of examples**.

**That is Machine Learning.** A machine, like Ramesh, looking at thousands of past examples and figuring out the pattern **by itself** — so it can predict the future.

## The big shift: old way vs ML way

This one picture is the entire reason ML exists:

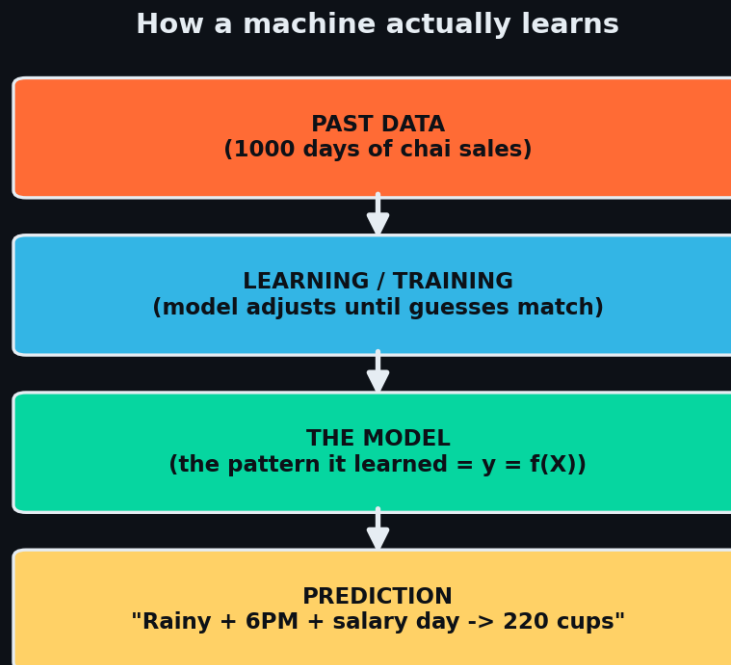


In normal coding YOU write the rules. In ML you give examples and the machine writes the rules itself.

**The intuition:** In normal coding, *you* are the smart one — you write every rule. In ML, you let the **machine become smart** by showing it examples.

**Why this is revolutionary:** some problems have too many rules to ever write by hand. How would you write if-else rules to spot a cat in a photo? Impossible. But show a machine 10,000 cat photos and it learns "cat-ness" on its own.

## How learning actually happens



Data → Training → Model → Prediction. The "model" is just the pattern it learned.

## You already use ML 20 times a day

| When you...                       | ML is quietly working           |
|-----------------------------------|---------------------------------|
| Open YouTube → perfect next video | Learned your watch history      |
| GPay/PhonePe flags a fraud txn    | Learned what fraud looks like   |
| Instagram reels know you too well | Learned your scroll patterns    |
| Gmail auto-sorts spam             | Learned from billions of emails |

## The Math (technical — skip if non-technical, you lose nothing)

Every ML model learns a **function**  $f$  that maps input  $X$  to output  $y$ :

$$y = f(x)$$

$x$  = the inputs (weather, time, crowd) -> "features"  
 $y$  = the answer (cups of chai sold) -> "label"  
 $f$  = the pattern (what the machine learns)-> the "model"

Training = adjusting  $f$  to shrink the gap between its guess and the truth. That gap is the **loss**. All of training is: "make the loss smaller."

## Try it — your first model in 6 lines

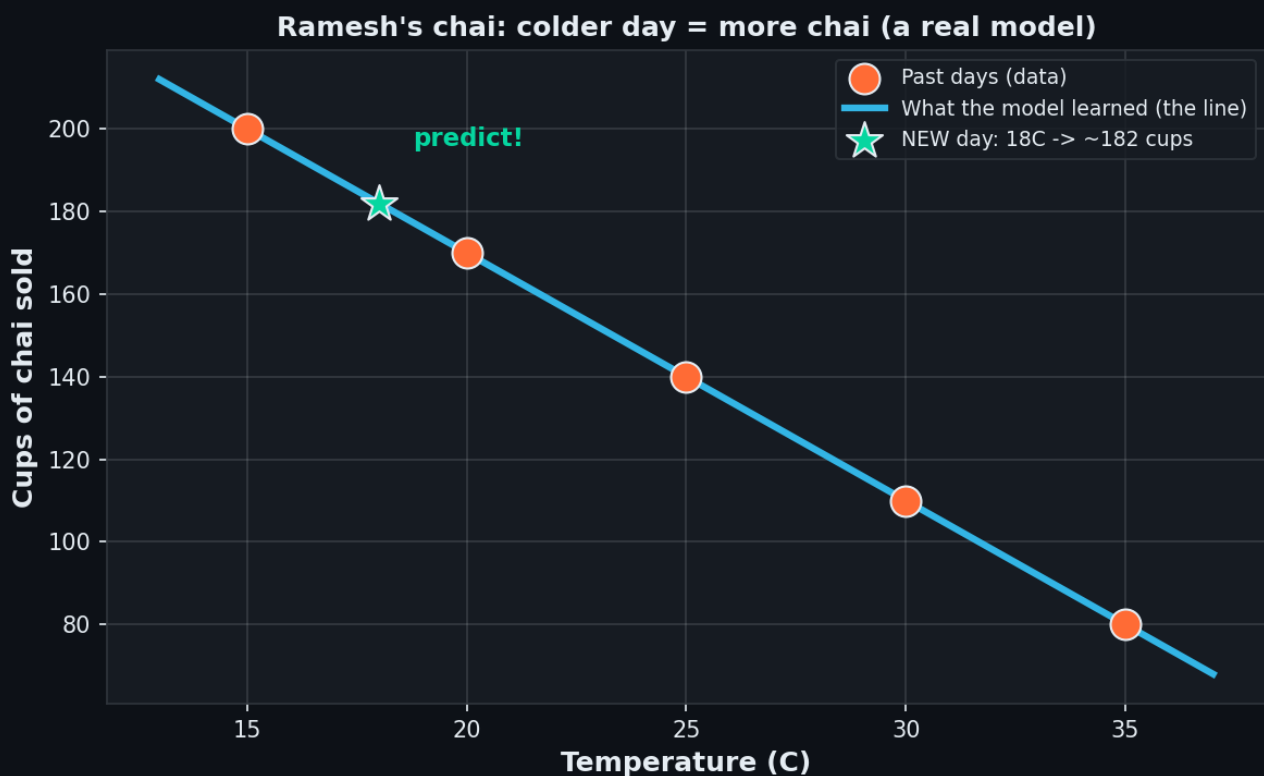
```

from sklearn.linear_model import LinearRegression

X = [[15],[20],[25],[30],[35]] # temperature
y = [200,170,140,110,80]      # cups sold

model = LinearRegression()
model.fit(X, y)                # <- THIS is learning

print(model.predict([[18]]))   # 18C -> ~182 cups
  
```



**`model.fit(X, y)`** is the entire heart of Machine Learning. Everything in the next 29 days is a more powerful version of it.

## Recap — the 20-second version

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- ML = learning patterns from examples, not writing rules by hand.
- Old way: you write rules. ML way: the machine finds the rules.
- The learned pattern is called a model ( $y = f(X)$ ).
- You already use it 20× a day — YouTube, GPay, Instagram.
- Training = making wrong guesses less wrong (shrinking the loss).

Next up — [Video 2](#): The 3 Families of ML: Supervised, Unsupervised, Reinforcement.  
Just like a child learns 3 ways — see you Day 2.